

PROJECT DOCUMENTATION

Project Title

Restaurant Business Analytics Dashboard using Dash

Project Overview

The Restaurant Business Analytics Dashboard is an interactive data visualization application developed to analyze customer behavior, revenue patterns, and staff performance using restaurant transaction data.

The dashboard enables business stakeholders to monitor key performance indicators (KPIs) and gain actionable insights into restaurant operations such as sales trends, tipping behavior, customer demographics, and dining patterns.

The system allows users to filter data dynamically by **day** and **time (Lunch/Dinner)** to perform focused analysis and make data-driven business decisions.

Business Objectives

This dashboard helps answer critical business questions such as:

- What is the total revenue generated?
- How much tip is collected (staff performance)?
- What is the average tipping percentage?
- What is the average customer bill?
- Which day generates peak revenue?
- Do lunch or dinner sessions earn more?
- Do smokers vs non-smokers contribute more revenue?
- How does customer group size affect revenue?
- Are there spending differences across gender?

KPIs Implemented

1. **Total Revenue** – Overall sales performance
2. **Total Tips Collected** – Staff service performance
3. **Average Tip Percentage** – Customer satisfaction indicator
4. **Average Bill Value** – Spending behavior
5. **Peak Revenue Day** – Best performing day

Visual Insights Provided

The dashboard includes the following analytical visualizations:

- Revenue by Day
- Revenue by Time (Lunch vs Dinner)
- Smoking vs Non-Smoking Revenue
- Tip Percentage Distribution
- Total Bill vs Tip Relationship
- Revenue by Gender
- Revenue by Group Size
- Day + Time Revenue Comparison

Key Features

- Interactive Multi-Select Filters
- Single Screen Business View
- Real-Time KPI Updates
- Customer Behaviour Analysis
- Operational Performance Monitoring

Tools & Libraries Used

Tool	Purpose
Python	Core programming
Dash	Interactive dashboard framework
Plotly Express	Data visualization
Pandas	Data manipulation
Seaborn	Dataset source
HTML Components	Layout design

Dataset Used

The project uses the **Tips Dataset**, which contains restaurant transaction data including:

- Total Bill Amount
- Tip Given
- Customer Gender
- Smoking Preference
- Day of Visit

- Dining Time (Lunch/Dinner)
- Group Size

Outcome

The dashboard provides a centralized analytical view of restaurant performance, enabling data-driven decision-making in areas such as staffing, marketing strategy, customer segmentation, and service quality improvement.

OUTPUT :



